

SAMANALI JAYARATHNA

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Career Profile

Data Science graduate from RMIT University with hands-on experience developing end-to-end machine learning pipelines, building AI-driven applications and analysing large-scale datasets. Proficient in Python, SQL, and ML frameworks including scikit-learn and FAISS, with experience in generative AI, RAG systems and LLM integration. Strong foundation in responsible AI practices, cloud technologies and software engineering best practices including GitHub. Experienced in communicating analytical findings clearly to both technical and non-technical stakeholders.

Education

Bachelor of Computer Science (Honours), RMIT University

July 2026 - July 2027

Bachelor of Data Science, RMIT University.

July 2023 - July 2026

- GPA: 3.5/4.0 (Pass with Distinction)
- Executive Committee member of RMIT Sri Lanka Association
- General Committee in RMIT SWITCH (Society for Women in Information Technology)

Foundation in Engineering, Swinburne University of Technology

July 2022 - June 2023

- Completed a Foundation Year in Engineering

Certifications -

RMIT Plus Program- RMIT University

Feb 2026

edX (AWS-DA-02), Getting started with Data Analytics on AWS

Dec 2024

edX, Data Science: R Basics

Nov 2023

Work Experiences

Content Creator (Part-time), ICS Migration and Education

Aug 2026 - Present

- Plan, film and edit short-form video content for the business's social media pages, taking each piece from concept through to final cut on a regular posting schedule.
- Act as the on-camera face for a new product launch, presenting to camera and adapting delivery based on feedback from the team and client response.
- Translate migration and education service information into accessible, engaging content for a broad audience, balancing accuracy with approachability.

Trainee - SAW Engineering Pvt Ltd

Nov 2025 - Feb 2026

- Contributed to the environmental sustainability division, supporting green technology initiatives including food waste composting and solar panel projects, assisting with project management activities and gaining hands-on exposure to sustainable engineering and clean energy implementation across Sri Lanka.
- Conducted data analysis to support operational and business development decisions, synthesising product specifications and cost data into structured reports for internal stakeholders
- Represented the company at industry exhibitions across Sri Lanka, engaging directly with clients and industry professionals to promote the company's product range and build professional relationships

Volunteering

Event Coordinator – RMIT SWITCH

March 2026 - Present

- Organised end-to-end logistics for tech-focused events including networking sessions, coding competitions, and industry collaborations with partners such as Notion.

Peer Mentor – Machine Learning, RMIT University

Feb 2026 - Present

- Mentored students on model evaluation, overfitting, and implementation trade-offs, bridging theoretical concepts with practical Python application.

Publicity Officer - RMIT Sri Lankan Association

August 2024 – August 2026

- Managed the association's social media presence and led publicity for large-scale cultural events attracting hundreds of attendees

Retail Assistant - Australian Red Cross, Richmond Bridge Rd

February 2025 – March 2026

- Provided customer service and supported daily retail operations including stock management alongside a team of volunteers.

University Projects

Eunoia: RAG-Powered Mental Health Chatbot | GitHub Repo: <https://github.com/Samanali-J/Eunoia>

- Designed and implemented an evaluation framework to measure retrieval quality and response reliability, using prompt grounding as a guardrail to constrain model outputs and reduce hallucination risk.
- Engineered a dual-index retrieval architecture searching a knowledge collection and Q&A bank in parallel, improving context relevance for downstream response generation
- Implemented a prompt-grounding guardrail layer to constrain model outputs to retrieved context, reducing hallucination risk and ensuring reliable, safe responses
- Delivered via an iterative agile workflow - contributing to sprint planning, daily standups, and managed version control through feature branches and pull requests on GitHub.

Key Technologies - Python, RAG, FAISS, Ollama, LLMs, Flask, sentence embeddings

Donation Site Operational Hours Monitoring - Power BI Dashboard & Python Preprocessing Pipeline | RMIT WIL

- Engineered a Python pipeline to extract, clean, and reconcile alarm schedules against donor centre operating hours across 82 fixed Australian Red Cross Lifeblood sites, identifying scheduling discrepancies at scale
- Leveraged the Claude Vision API to automate extraction of structured schedule data from complex multi-page PDFs, replacing a manual validation process that was not feasible across hundreds of pages
- Built a reconciliation engine that compares expected alarm activation times against actual site hours using configurable business rules, flagging mismatches, missing schedules, and internal scheduling conflicts across all site-day combinations
- Delivered a Power BI dashboard with KPI cards, state and day-level breakdowns, and a site deep dive page, enabling the operations team to identify and resolve alarm scheduling errors across the national network

Key Technologies - Python, Pandas, Claude Vision API (Anthropic), PyMuPDF, pdfplumber, Power BI, Power Query

Visual Entailment Classification: Multimodal Deep Learning System | RMIT

- Built and iteratively evaluated three multimodal architectures (ResNet50 + BERT fusion models) for a visual entailment task, progressing from concatenation-based fusion to cross-attention and self-attention mechanisms
- Applied stratified image-level train, validation, and test splitting to prevent data leakage across correlated samples, and validated model selection against accuracy, AUC, precision, and recall
- Conducted a documented hyperparameter search across learning rate, dropout, and layer-freezing strategies, improving validation AUC from 0.71 to 0.82 through iterative fine-tuning
- Achieved 72% test accuracy and 0.80 AUC on held-out data, demonstrating generalisation beyond the training set

Key Technologies: Python, TensorFlow, Keras, BERT, ResNet50, scikit-learn

Skills

Technical:

- **Programming Languages:** Python, R, SQL
- **Data Tools:** Pandas, NumPy, Matplotlib, seaborn, Plotly, scikit-learn, , TensorFlow, Keras, Excel, PowerBI
- **Databases:** MySQL, MongoDB
- **Cloud & Tools:** AWS (EC2, S3, Lambda, ECS), Docker, GitHub, Google Colab, Flask
- **Engineering Practices:** ETL/ELT pipeline development, Agile delivery (sprint planning, Kanban), version control(GitHub)
- **Security & Forensics Tools:** Autopsy, CISCO

Other: Microsoft Office Suite

Soft Skills:

- **Problem Solving:** Designed a hallucination guard solution mid-project when the AI system was generating unreliable outputs, finding a technical fix under time pressure
- **Critical Thinking:** Identified and implemented a flexible rule-based adjustment layer in the Red Cross data pipeline when client requirements changed midway through the project
- **Communication:** Regularly present data findings and project updates to both technical teammates and non-technical stakeholders across university and industry projects
- **Team Collaboration and Leadership:** Led event planning and on the day operations across two student organisations simultaneously while coordinating with diverse teams of people
- **Adaptability:** Rapidly self-taught RAG systems, vector databases, FAISS indexing, and LLM integration to deliver the Eunoia project beyond course requirements, applying new technologies within tight sprint deadlines

Languages: English (Fluent), Sinhala (Native)

Referees

Referees available upon request